

Homework 2

EN.601.439/639 | Fall 2026

You must submit either a physical handwritten homework or print out your homework to submit at the beginning of class.

Problem 1: VAEs with different data modalities

Recall that a variational autoencoder (VAE) is trained by maximizing the evidence lower bound objective (ELBO):

$$\operatorname{argmax}_{\theta, \phi} \sum_{i=1}^N \left[\mathbb{E}_{z_i \sim q_{\phi}(\cdot | x_i)} [\log p_{\theta}(x_i | z_i)] - \text{KL}(q_{\phi}(z_i | x_i) || p(z_i)) \right] \quad (1)$$

where:

- $p(z)$ is called the *prior* distribution;
- $p_{\theta}(x | z)$ is called the *likelihood* distribution and describes the distribution of the observed data x given its embedding z ; and
- $q_{\phi}(z | x)$ is called the *variational posterior* distribution, which approximates the posterior distribution $p(z | x)$ that describes the distribution of the embedding z given the observed data x .

In the standard VAE framework, these three distributions are all parametrized by normal distributions. In single cell data analysis, we may wish to modify these distributional assumptions depending on the type of data we are analyzing. In this question, we will derive different VAE optimization problems for different types of data.

- a) Suppose each data vector $x_i \in \mathbb{R}^G$ contains continuous measurements, such as normalized gene/protein expression values. Assume that the likelihood $p_{\theta}(x | z)$ is a Gaussian distribution, and that the other distributions $p(z), q_{\phi}(z | x)$ have the same form as in a standard VAE.
 - Write out the equation for the likelihood distribution $p_{\theta}(x | z)$. What are the learned parameter(s) θ ?
 - Write out the VAE optimization problem (1).
- b) Next, replace the standard Gaussian prior $p(z)$ with a Student's t -distribution, which has “heavier tails” than a Gaussian.
 - Write out the equation for the prior distribution $p(z)$.
 - Write out the VAE optimization problem (1).

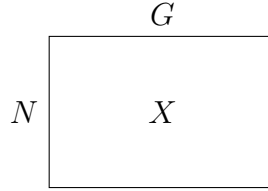
- How might the using such a “heavy-tailed” prior $p(z)$ change the learned distributions $p_\theta(x | z)$ and/or $q_\phi(z | x)$ compared to using a standard Gaussian?
- c) Suppose $x_i \in \{0, 1\}^L$ contains binary measurements, e.g. chromatin accessibility at L different genetic loci. Assume that the likelihood $p_\theta(x | z)$ is a Bernoulli distribution and that the other distributions $p(z), q_\phi(z | x)$ have the same form as in a standard VAE.
- Write out the equation for the likelihood distribution $p_\theta(x | z)$. What are the learned parameter(s) θ ?
 - Write out the VAE optimization problem (1).
- d) Suppose $x_i \in \mathbb{R}^G$ contains nonnegative count measurements, such as UMI counts for G different genes. We may model the likelihood distribution $p_\theta(x_i | z_i)$ for such count measurements with either a Poisson distribution or a negative binomial distribution.
- Write out the equation for the likelihood distribution $p_\theta(x | z)$ for each choice of distributions. What are the learned parameter(s) θ ?
 - Write out the VAE optimization problem (1) for each choice.
 - Why would you use a negative binomial distribution over a Poisson distribution?
- e) Using Pyro and a coding agent of your choice, implement VAEs for one of the likelihood distributions above and train your VAE on a single cell or ST dataset. Submit a code notebook along with your homework. In your notebook, please do the following:
- State the likelihood distribution and its parameterization.
 - Plot the ELBO versus epoch.
 - Visualize the learned embedding z .
 - Add a brief discussion on whether the likelihood distribution $p_\theta(x | z)$ is appropriate for the data you analyzed, and whether you think the learned embedding z is biologically meaningful.

Problem 2: Linear autoencoder is PCA

Let $x_1, \dots, x_N \in \mathbb{R}^{1 \times G}$ be (mean-centered) gene expression vectors, where N is the number of cells and G is the number of genes. (For this question, we will represent each individual gene expression vector x_i as a *row* vector.) We arrange these observations as the rows of a single cell gene expression matrix $X \in \mathbb{R}^{N \times G}$.

We will show that training a *linear autoencoder*, i.e. an autoencoder with no activation functions or bias terms, trained with a mean squared error (MSE) loss, is equivalent to a specific formulation of PCA.

Tip: It may be helpful to write out matrices using rectangular diagrams; e.g. you can write the data matrix X as:



- a) Let $f_\theta : \mathbb{R}^G \rightarrow \mathbb{R}^K$ be the encoder function and $g_\phi : \mathbb{R}^K \rightarrow \mathbb{R}^G$ be the decoder function for an autoencoder with parameters θ, ϕ , respectively. Write out the optimization problem to learn the encoder and decoder functions from the N gene expression vectors $x_1, \dots, x_N$ with a squared error loss. Your objective function should include a summation over $i = 1, \dots, N$. (*Hint: see class notes.*)
- b) Next, assume the encoder and decoder are linear functions of the form $f(x) = xA$ and $g(z) = zB$, where we represent $x \in \mathbb{R}^{1 \times G}$ and $z \in \mathbb{R}^{1 \times K}$ as row vectors, and the matrices $A \in \mathbb{R}^{G \times K}$ and $B \in \mathbb{R}^{K \times G}$ are (learned) matrices. Rewrite the optimization problem from part a using these linear functions.
- c) Rewrite the optimization problem from part b in terms of matrices X, A , and B , rather than as a summation. (*Hint: use the Frobenius norm $\|M\|_F = \sqrt{\sum_{i,j} M_{ij}^2}$, or the square-root of the sum of squares of all entries of a matrix M .*)
- d) Show that the optimization problem from part c is equivalent to the following optimization problem:

$$\min_{\substack{C \in \mathbb{R}^{G \times G} \\ \text{rank}(C) \leq K}} \|X - XC\|_F^2. \quad (2)$$

- e) Let $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_{\min(N,G)}$ be the singular values of X ; assume that $\sigma_K \neq \sigma_{K+1}$ and $K < \min(N, G)$. Show that the optimization problem in Equation (2) is equivalent to the following optimization problem

$$\min_{\substack{Y \in \mathbb{R}^{N \times G} \\ \text{rank}(Y) \leq K}} \|X - Y\|_F^2, \quad (3)$$

in the sense that any solution Y^* to (3) has the form $Y^* = XC^*$ for some C^* with $\text{rank}(C^*) \leq K$.

Hint 1: Letting $X = U\Sigma V^T$ be the singular value decomposition of X , then the unique solution Y^ to Eq. (3) has the form $Y^* = U_K \Sigma_K V_K^T$ where U_K are the first K columns of U ; Σ_K is the top $K \times K$ block of Σ , and V_K^T are the first K rows of V^T . Now you must find a matrix $C \in \mathbb{R}^{G \times G}$ with $\text{rank}(C) \leq K$ such that $XC = U_K \Sigma_K V_K^T$.*

Hint 2: Try constructing C using U_K and/or V_K .